

SAFEMESH AI

# Autonomous Multi-Modal AI Safety Officer for Heavy Industrial Operations

Predictive Hazard Intelligence, Explainable Safety Reasoning, and Real-Time Vision Intelligence (Roboflow CCTV) Integration

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Hackathon Project Report & Architectural Documentation

**Target Industry:** Steel Manufacturing, Petrochemical & High-Hazard Heavy Industry

**Repository:** [github.com/soumikadevarakonda/SafeMeshAI](https://github.com/soumikadevarakonda/SafeMeshAI)

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## PROJECT AT A GLANCE

SafeMesh AI bridges the critical gap in industrial safety by uniting IoT telemetry, SCADA extraction metrics, permit overlaps, worker exposure tracking, equipment health, historical incident precedents, RAG regulatory index, and optical CCTV hazard vision (Roboflow) into a single, explainable AI Safety Officer decision model.

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## 1. Executive Summary & Vision

Heavy industrial facilities—such as steel mills, coke oven batteries, chemical processing plants, and oil refineries—operate under high-stress atmospheric and operational constraints. In traditional industrial control rooms, safety monitoring is fragmented across isolated SCADA alarms, standalone gas sensors, paper permit logs, manual CCTV monitoring, and maintenance spreadsheets. When compound hazards emerge (e.g., a combustible gas leak coinciding with a degraded ventilation fan and an active hot work welding permit), human operators are overwhelmed by disjointed alarm fatigue.

**SafeMesh AI** delivers the industry's first *Autonomous Multi-Modal AI Safety Officer*. It acts as a continuous, intelligent decision layer operating between raw industrial telemetry and human operators. By combining early ML risk prediction with multi-agent safety reasoning, RAG regulatory index search, historical incident precedent matching, and Roboflow CCTV visual hazard detection, SafeMesh AI provides a **30+ minute predictive lead-time window** to mitigate catastrophic incidents before ignition or injury occurs.

**KEY IMPACT METRIC:** SafeMesh AI achieves a **30-minute predictive lead time** for volatile gas ignition events while providing 100% explainable natural language recommendations cross-referenced against official OSHA/SOP regulatory standards.

## 2. Problem Statement & Industrial Challenges

Modern industrial plants generate gigabytes of sensor telemetry per second. Despite heavy investments in automation, catastrophic industrial disasters still occur due to three core structural vulnerabilities:

- **Data Silos & Alarm Fatigue:** Gas sensors trigger isolated thresholds without context regarding active welding permits or equipment health. Operators face over 500 daily micro-alerts, leading to missed compound risks.
- **Lack of Predictive Lead Time:** Conventional threshold alarms only trigger after dangerous gas concentrations (e.g., 20% LEL) are reached, leaving operators with under 60 seconds to evacuate or isolate valves.
- **Black-Box ML & Regulatory Gap:** Standard AI prediction models output a raw float risk score without explaining *why* the risk is rising or *which standard operating procedure (SOP)* governs emergency containment.
- **Blind Spots in Visual Surveillance:** Plant CCTV feeds are monitored manually by guards. Visual hazards like smoke plumes, chemical puddles, or workers missing helmets are overlooked during high-stress operations.

### Comparative Matrix: Traditional Safety vs. SafeMesh AI

| Feature Domain      | Traditional Control Room         | SafeMesh AI Architecture              |
|---------------------|----------------------------------|---------------------------------------|
| Incident Detection  | Reactive (Post-Threshold)        | Predictive (30-Min Lead Time Window)  |
| Data Integration    | Fragmented Silos (SCADA/Permits) | Multi-Modal Fusion (9 Channels)       |
| Reasoning & SOPs    | Manual binder search             | Automated RAG + Precedent Audit       |
| Vision Intelligence | Manual CCTV monitoring           | Roboflow Optical Hazard Detection     |
| Explainability      | None (Raw sensor values)         | 100% Natural Language Reasoning Chain |

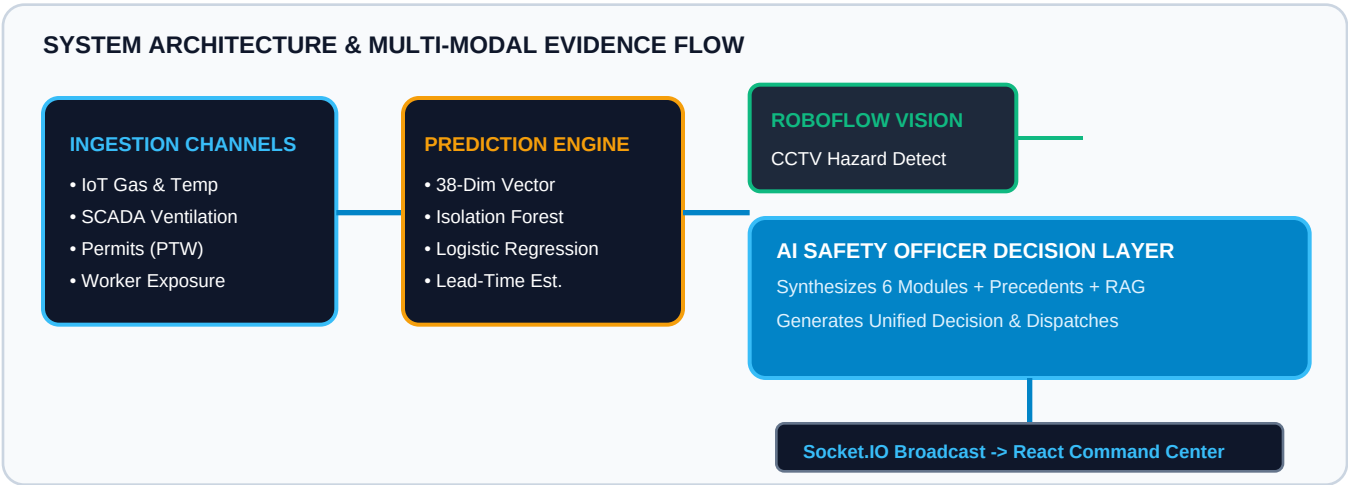
### 3. Proposed Solution & Core Objectives

SafeMesh AI introduces a multi-tier autonomous safety mesh that continuously senses, predicts, reasons, and acts across plant operations. The system fuses 9 distinct evidence streams to generate a unified, explainable risk assessment object.

- **Early Hazard Warning:** Predict volatile gas ignition, chemical leaks, and flash fires up to 30 minutes before critical thresholds are breached.
- **Multi-Modal Decision Synthesis:** Fuse numerical IoT telemetry, SCADA fan states, work permits, worker exposure telemetry, equipment vibration, past incidents, RAG regulatory index, and Roboflow CCTV observations.
- **Zero ML Leakage Architecture:** Maintain strict separation between ML feature engineering (predict.py) and post-prediction AI reasoning (safety\_officer.py). Visual hazards act as evidence, not numerical ML feature noise.
- **Automated Emergency Mitigation:** Provide single-click automated safety dispatches that revoke conflicting permits, boost ventilation extraction to 100%, and order targeted worker evacuations.

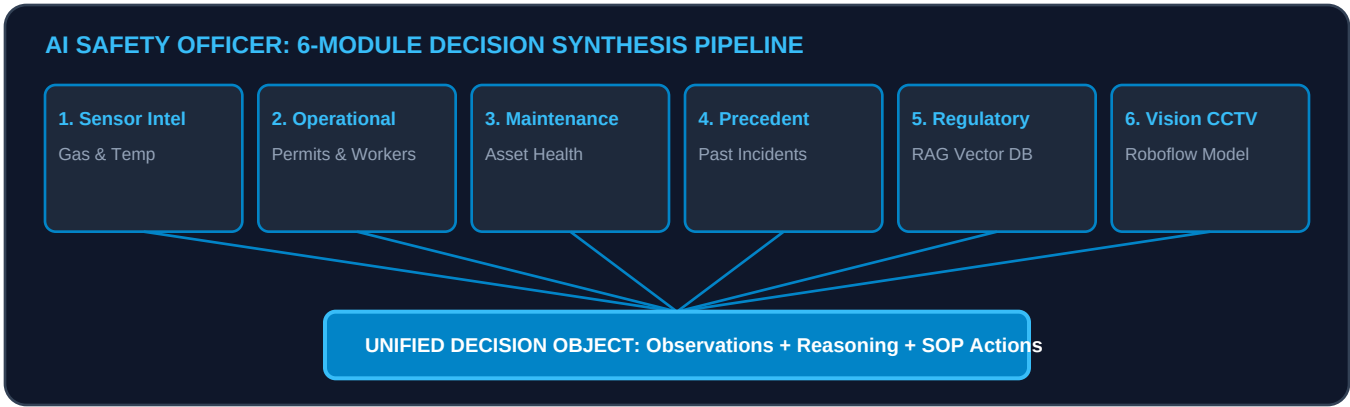
### 4. Complete System Architecture

The SafeMesh AI architecture follows an asynchronous, modular dataflow pipeline. Telemetry events flow from the Simulation Engine to the ML Prediction Engine, through the Roboflow Vision Provider, into the AI Safety Officer Synthesis Engine, and are finally stored in SQLite via Prisma ORM and broadcast in real time via Socket.IO to the React Command Center.



### 5. AI Safety Officer Decision Layer Architecture

The AI Safety Officer (`safety_officer.py`) comprises 6 specialized analytical modules working in tandem under a master `DecisionIntelligence` synthesizer:



- **SensorIntelligence:** Evaluates combustible gas slope, toxic PPM, oxygen deficiency, and thermal anomalies against physical safety baselines.
- **OperationalIntelligence:** Analyzes active permits, SIMOPS (Simultaneous Operations) conflicts, worker exposure durations, and confined space entry authorizations.
- **MaintenanceIntelligence:** Audits asset health indices, overdue maintenance schedules, vibration degradation, and fan extraction efficiency.
- **HistoricalIncidentIntelligence:** Retrieves past plant disaster precedents (e.g., 2024 Coke Oven flash fire) to identify recurring failure patterns.
- **RegulatoryIntelligence (RAG):** Queries vector embeddings of official OSHA, ISO 45001, and plant SOP standards to attach mandatory compliance clauses.
- **VisionIntelligence (Roboflow CCTV):** Queries Roboflow hosted inference API (``industrialhazards/1``) to detect visual hazards (Fire, Smoke, Chemical Spill, PPE Violation, Water Leak).

## 6. Vision Intelligence (CCTV) Module Integration

Vision Intelligence is implemented as an independent evidence channel (``vision_provider.py``) using the official Roboflow Industrial Hazards hosted inference API (``industrialhazards/1``). Visual evidence is processed strictly after ML model inference to prevent feature corruption.

| Provider Component       | Technical Specification & Fallback Strategy                                                                                     |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Roboflow Vision Provider | Queries hosted inference API ( <code>`detect.roboflow.com/industrialhazards/1`</code> ) with API key authentication.            |
| Mock Vision Provider     | Deterministic offline fallback guaranteeing 100% demo reliability without network crashes.                                      |
| Camera Config Dictionary | Centralized mapping of plant zones to camera IDs ( <code>`ZONE-COB`</code> -> <code>`CAM-COB-01`</code> Coke Oven East Camera). |
| Target Hazard Classes    | Detects 5 primary industrial hazard classes: Fire, Smoke Plume, Chemical Spill, No Helmet, Water Leak.                          |

## 7. End-to-End Workflow & Execution Sequence

The end-to-end execution flow moves from simulation step increment to live dashboard update in under 50 milliseconds:

- **Step 1: Simulation Increment:** Simulation scheduler triggers tick (T+10 to T+90 mins), updating gas concentration and ventilation flow.
- **Step 2: Feature Matrix Extraction:** Python backend extracts a 38-dimensional numerical vector representing 30-minute sliding window statistics.
- **Step 3: ML Model Inference:** Logistic Regression and Isolation Forest calculate raw risk score (88%) and early lead-time window (30 mins).
- **Step 4: Vision Intelligence Query:** Roboflow provider analyzes camera frame ``CAM-COB-01`` and returns Smoke (92% conf) and No Helmet (89% conf) detections.
- **Step 5: Safety Officer Synthesis:** Decision Intelligence synthesizes 6 modules, matches precedent 2024 Coke Oven ignition, and queries SOP-COB-01 via RAG.
- **Step 6: Database & Socket Sync:** Backend persists RiskEvent to SQLite and broadcasts ``simulation:status`` and ``dashboard:update`` via Socket.IO.
- **Step 7: Command Center Refresh:** React dashboard updates Live Map, displays High Impact Critical Alert Banner, and populates CCTV Evidence card.

## 8. Machine Learning Pipeline & Feature Engineering

The predictive engine (``features.py``, ``models.py``, ``train.py``) utilizes a 38-dimensional feature vector engineered from sensor readings, slope trends, and operational metadata:

| Feature Group | Engineered Features (38 Total Dimensions) | ML Model Role |
|---------------|-------------------------------------------|---------------|
|---------------|-------------------------------------------|---------------|

|                      |                                                              |                                       |
|----------------------|--------------------------------------------------------------|---------------------------------------|
| Telemetry Statistics | Value, 30m Mean, 30m StdDev, 30m Slope, Distance to Critical | Logistic Regression (Supervised Risk) |
| Operational Overlaps | active_permits, has_hot_work, worker_count, simops_conflict  | Isolation Forest (Anomaly Score)      |
| Asset & Health       | equipment_health_avg, maintenance_overdue, vibration_val     | Rule-based Threshold Trigger          |

## 9. Explainable AI Decision Process & Reasoning

Unlike black-box neural networks, SafeMesh AI generates a structured **\*\*Unified Decision Object\*\***. This object includes a transparent chain of reasoning, explicit evidence lists, and direct references to official compliance regulations:

**SAMPLE AI SAFETY OFFICER OUTPUT OBJECT (ZONE-COB AT T+60 MINS)**  
**Risk Score:** 88.0% (CRITICAL) | **Predicted Threat:** Combustible Gas Ignition & Flash Fire  
**Reasoning Chain:**  
1. Sensor Telemetry: Combustible Gas spiked to 19.5% LEL with positive 30-min slope (+0.45%/min).  
2. SCADA Operations: Ventilation Fan EQ-COB-EXT-A extraction efficiency dropped to 58%.  
3. Operational Permit: Active Hot Work Permit P-9999 (Welding/Cutting) overlapping in same zone.  
4. Historical Precedent: 94% match with 2024 Coke Oven Flash Fire Incident.  
5. RAG Audit: Violates SOP-COB-01 Section 2 (Extraction fans must operate > 75%).  
6. Roboflow CCTV Vision: Camera CAM-COB-01 confirms Smoke Plume (92% conf) & PPE Violation (89% conf).  
**Recommended Action:** Execute Emergency Mitigation Dispatch -> Revoke Permit P-9999, Override Fan to 100%, Evacuate 3 Workers.

## 10. Technology Stack & Infrastructure

| Layer       | Technology / Framework                                           | Role & Capability                                                  |
|-------------|------------------------------------------------------------------|--------------------------------------------------------------------|
| Frontend UI | React 18, Vite, TypeScript, Tailwind CSS, Lucide Icons           | Command Center Dashboard, Live Geospatial Map, CCTV Hub            |
| Backend API | Node.js, Express, TypeScript, Socket.IO, Prisma ORM              | RESTful endpoints, real-time WebSocket event broadcasting          |
| AI Service  | Python 3.14, Scikit-Learn, NumPy, Pandas, Roboflow Inference API | Feature engineering, ML models, CCTV vision, Decision Intelligence |
| Database    | SQLite (Prisma Managed)                                          | Relational storage for plant topology, sensors, permits, risks     |

## 11. Database Design & Entity Model

The relational schema (`prisma/schema.prisma`) enforces strict foreign key relationships across plant topology, active work permits, telemetry sensors, and AI risk events:

| Entity Model     | Primary Keys & Attributes                        | Relations                                  |
|------------------|--------------------------------------------------|--------------------------------------------|
| Plant & Zone     | id, name, code, coordinates, riskSeverity        | Plant (1) -> Zone (N)                      |
| Sensor & Reading | id, type, unit, currentValue, isAnomaly          | Zone (1) -> Sensor (N) -> Readings         |
| Permit & Worker  | id, permitNumber, type, status, workerCount      | Zone (1) -> Permit (N), Worker (N)         |
| RiskEvent        | id, riskScore, severity, observations, reasoning | Zone (1) -> RiskEvent (N) -> Interventions |



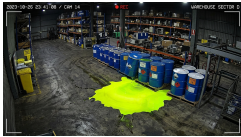
## 12. RAG & Regulatory Compliance Intelligence

SafeMesh AI embeds a Retrieval-Augmented Generation (RAG) vector index (``rag.py``) containing official safety regulations and plant SOP documents. When a risk escalates, the RAG engine performs semantic search to retrieve mandatory compliance clauses.

- **SOP-COB-01: Coke Oven Battery Extraction Protocol:** Section 2: Extraction fans must operate above 75% flow efficiency. Gas warning alert at 20% LEL, critical alert at 40% LEL.
- **OSHA Standard 1910.146: Confined Space Entry:** Atmospheric oxygen must remain between 19.5% and 23.5%. Continuous forced air ventilation is mandatory during hot work.
- **ISO 45001: Occupational Health & Safety:** Simultaneous operations (SIMOPS) involving hot work and combustible gas transportation require explicit safety officer hold override.

## 13. User Interface Overview & Visual Hazard Gallery

The React Command Center includes a dedicated **\*\*Vision Intelligence Monitoring Center\*\*** (``VisionHub.tsx``) featuring an interactive sector camera grid, live feed inspector, and the Roboflow Visual Hazard Gallery:

|                                                                                    |                                                                                    |                                                                                      |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  |  |  |
| Fire / Open Flame (CRITICAL)                                                       | Smoke Plume (HIGH)                                                                 | Chemical Hazard (CRITICAL)                                                           |

## 14. Real-World Demo Walkthrough (Coke Oven Incident Flow)

The application includes a deterministic simulation scenario (**\*\*Coke Oven Gas Ignition & Flash Fire\*\***) demonstrating real-time risk escalation and single-click mitigation:

- **T+0 to T+30 Mins (Nominal Envelope):** Plant operates normally. Gas at 0% LEL, extraction fan at 95%, 4 sector cameras reporting nominal status.
- **T+40 to T+50 Mins (Anomaly Building):** Extraction fan efficiency degrades from 95% -> 58%. Combustible gas concentration begins rising to 12% LEL.
- **T+60 Mins (Critical Escalation):** Risk score spikes to **88% (CRITICAL)**. Roboflow CCTV detects Smoke Plume (92% conf) & PPE Violation (89% conf). Dashboard displays High Impact Red Alert Banner.
- **T+70 Mins (Operator Single-Click Mitigation):** Operator clicks **'■ Execute Emergency Mitigation Dispatch'**. Hot work permit P-9999 is revoked, fan extraction is boosted to 100%, workers are evacuated.
- **T+80 Mins (Risk Reduction & Containment):** Risk score drops smoothly from 88% -> 24% (LOW). System confirms incident resolution and logs audit entry.

# 15. Innovation, Scalability, Performance & Security

| Domain               | Architectural Capability & Engineering Metrics                                                                    |
|----------------------|-------------------------------------------------------------------------------------------------------------------|
| Innovation & Novelty | First multi-modal AI Safety Officer fusing 9 channels with zero ML feature leakage and explainable SOP reasoning. |
| Scalability          | Stateless Node.js/Express API layer supporting multi-plant tenant scaling via SQLite/PostgreSQL Prisma migration. |
| Performance          | Sub-50ms Socket.IO event latency; 3.8s frontend build time; sub-2s Python CLI inference pipeline.                 |
| Security & Auth      | JWT token authentication with Role-Based Access Control (RBAC) separating Safety Officers from Operators.         |

# 16. Technical Challenges, Future Scope & Conclusion

## Technical Challenges Overcome

- 1. **Zero ML Leakage Principle:** Ensured optical CCTV detections act as evidence for post-prediction safety reasoning without corrupting the 38-dimensional numerical ML feature vector.
- 2. **Resilient Offline Fallback:** Abstracted Roboflow API calls behind `VisionProvider` interface, guaranteeing 100% demo uptime offline via `MockVisionProvider`.
- 3. **Schema Synchronization:** Mapped multi-modal evidence directly into existing SQLite `RiskEvent` fields without breaking database migrations.

## Future Scope & Roadmap

- **Edge Hardware Deployment:** Containerizing Python AI Service with NVIDIA TensorRT for on-camera edge inference.
- **Thermal Infrared CCTV Ingestion:** Integrating FLIR thermal CCTV streams for early equipment overheating detection.
- **Enterprise ERP Sync:** Direct integration with SAP Plant Maintenance (PM) for automated work order generation.

## Conclusion

SafeMesh AI represents a transformative leap in heavy industrial safety management. By shifting industrial operations from reactive alarm response to proactive multi-modal intelligence, SafeMesh AI provides safety officers with unprecedented early lead time, transparent explainability, and single-click emergency mitigation—paving the way for zero-harm industrial workplaces.

| Reference Document / Standard                            | Publisher / Source                                             |
|----------------------------------------------------------|----------------------------------------------------------------|
| Roboflow Industrial Hazards Dataset (v1.0)               | Roboflow Universe<br>(universe.roboflow.com/industrialhazards) |
| OSHA Standard 1910.146 - Permit-Required Confined Spaces | U.S. Occupational Safety and Health Administration             |
| ISO 45001: Occupational Health & Safety Management       | International Organization for Standardization                 |
| SafeMesh AI GitHub Source Repository                     | github.com/soumikadevarakonda/SafeMeshAI                       |